

1 Solution — GIAM 3.3

Problem 4

1.1 Problem

Prove (by contradiction) that there is no smallest positive real number.

1.2 Formalising the Claim

“ ε is the smallest positive real” means ε is itself positive and no positive real is below it. Writing $\mathbb{R}^+$ for the set of positive reals, the theorem is

$$\neg \left(\exists \varepsilon \in \mathbb{R}^+, \forall x \in \mathbb{R}^+, \varepsilon \leq x \right),$$

so the supposition to be contradicted is

$$\exists \varepsilon \in \mathbb{R}^+, \forall x \in \mathbb{R}^+, \varepsilon \leq x.$$

1.3 Answer

No such ε exists: $\varepsilon/2$ is a positive real below ε .

1.4 Proof

Theorem. *There is no smallest positive real number.*

Proof. *Suppose, to the contrary, that there is one; call it ε . Then $\varepsilon > 0$ and*

$$\forall x \in \mathbb{R}, \quad x > 0 \implies \varepsilon \leq x.$$

Consider $\varepsilon/2$. Since $\mathbb{R}$ is closed under division by the non-zero real 2 , this is a real number, and since $\varepsilon > 0$ and $2 > 0$ we have $\varepsilon/2 > 0$. So $\varepsilon/2$ is a legitim-

ate choice for x , giving

$$\varepsilon \leq \frac{\varepsilon}{2}.$$

Multiplying both sides by 2 (a positive number, so the inequality is preserved) yields $2\varepsilon \leq \varepsilon$, and subtracting ε from both sides gives $\varepsilon \leq 0$.

This contradicts $\varepsilon > 0$. Hence no smallest positive real number exists. ■

1.5 The Picture

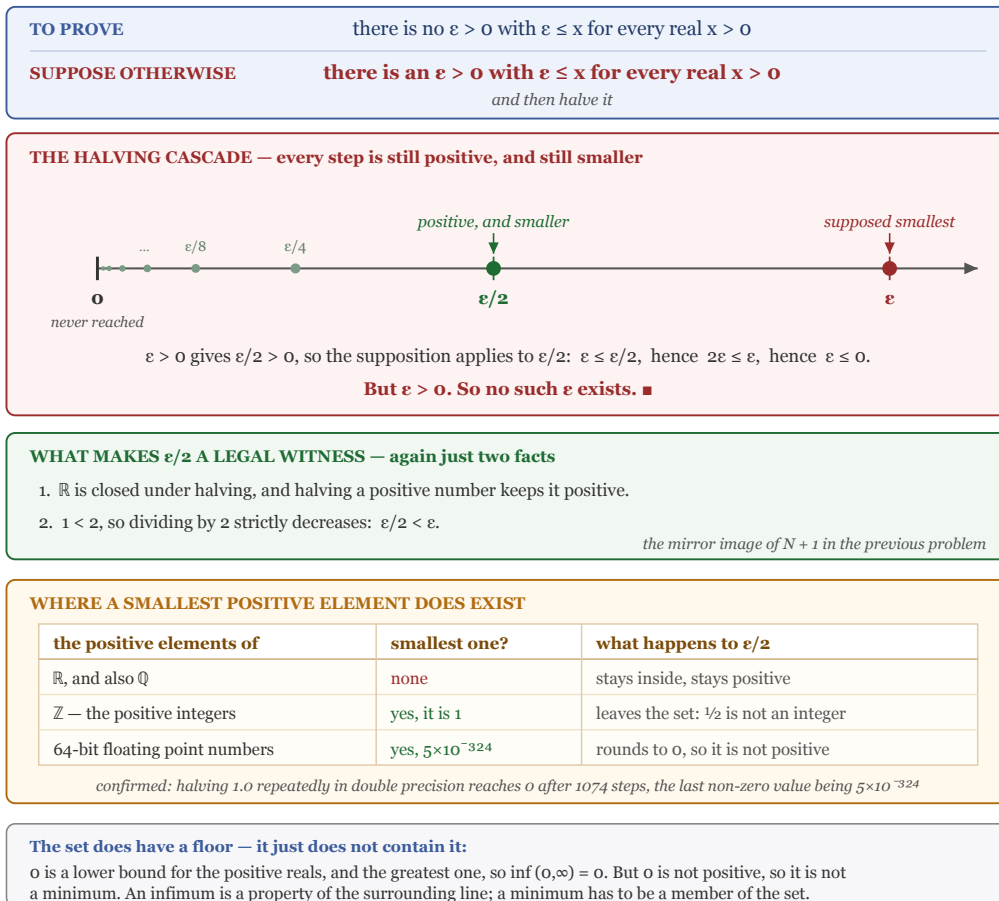


Figure 1.1: A diagram in five bands. The first states what is to be proved — there is no epsilon greater than zero with epsilon at most x for every real x greater than zero — then the supposition otherwise, that such an epsilon exists, and then: and then halve it. The second band shows a number line with 0 at the left marked ‘never reached’ and epsilon at the right marked ‘supposed smallest’; halfway between sits epsilon over 2, marked ‘positive, and smaller’, followed by progressively smaller dots at epsilon over 4, epsilon over

8 and beyond, piling up towards 0 without arriving. Below it: epsilon greater than zero gives epsilon over 2 greater than zero, so the supposition applies to epsilon over 2, giving epsilon at most epsilon over 2, hence 2 epsilon at most epsilon, hence epsilon at most 0; but epsilon is greater than zero, so no such epsilon exists. The third band lists what makes epsilon over 2 a legal witness, again just two facts: the reals are closed under halving and halving a positive number keeps it positive; and 1 is less than 2, so dividing by 2 strictly decreases. It is the mirror image of N plus 1 in the previous problem. The fourth band shows where a smallest positive element does exist: for the positive reals and also the positive rationals there is none, since epsilon over 2 stays inside and stays positive; for the positive integers the smallest is 1, because epsilon over 2 leaves the set as one half is not an integer; and for 64-bit floating point numbers there is a smallest, 5 times 10 to the minus 324, because epsilon over 2 rounds to 0 and so is not positive. Confirmed: halving 1.0 repeatedly in double precision reaches 0 after 1074 steps, the last non-zero value being 5 times 10 to the minus 324. The fifth band notes that the set does have a floor it does not contain: 0 is a lower bound for the positive reals and the greatest one, so the infimum of the open interval from 0 to infinity is 0, but 0 is not positive so it is not a minimum. An infimum is a property of the surrounding line; a minimum has to be a member of the set.

1.6 Remark — The Mirror of Problem 3, With One Change

The two problems have the same skeleton. Suppose the extreme element exists, produce a witness that beats it, collide. Only the witness differs:

	no largest integer	no smallest positive real
supposed extreme	N	ε
witness	$N + 1$	$\varepsilon/2$
operation used	add 1	divide by 2
why it stays in the set	$\mathbb{Z}$ closed under +	$\mathbb{R}^+$ closed under $\div 2$
why it beats the extreme	$0 < 1$	$1 < 2$

The change of operation is not cosmetic. Adding 1 would be useless here — $\varepsilon + 1$ is *bigger*, and $\varepsilon - 1$ may well be negative and so outside $\mathbb{R}^+$. What is needed is an operation that shrinks a positive number while keeping it positive, and multiplication by a factor strictly between 0 and 1 is exactly that. Halving is the obvious choice, but $\varepsilon/3$, ε/π , or $\frac{\varepsilon}{1 + \varepsilon}$ all serve.

One tempting witness that does **not** always work is ε^2 . It is smaller than ε only when $\varepsilon < 1$: for $\varepsilon = 2$ we get $\varepsilon^2 = 4$, which is larger. A witness has to work for *every* ε the supposition could hand you, not just the small ones you were picturing.

1.7 Remark — Where the Two Problems Come Apart

Problem 3 and this one look symmetric, but the underlying number systems are not. The *positive integers do have a smallest element* — it is 1 — and that is not an accident, it is the **well-ordering principle**: every non-empty set of positive integers has a least element.

So this pair of exercises quietly marks the boundary between the discrete and the continuous:

- $\mathbb{Z}$ has no largest element, and no smallest either, but every non-empty subset bounded below *does* have a minimum.
- $\mathbb{R}^+$ has no smallest element, and no non-empty open subset has one either.

The obstruction in $\mathbb{Z}$ is precisely that $\varepsilon/2$ escapes the set: $1/2$ is not an integer, so the halving argument dies at the first step. Between consecutive integers there is nothing; between any two distinct reals there are infinitely many more. That density is what this proof is really detecting.

1.8 Remark — Infimum Versus Minimum

The set $\mathbb{R}^+ = (0, \infty)$ is bounded below — every element exceeds 0 — so this is another instance of the distinction raised in Problem 3, now on the other side:

$$\inf(0, \infty) = 0, \quad \text{but} \quad \min(0, \infty) \text{ does not exist.}$$

The infimum is the greatest lower bound, and it need not belong to the set; the minimum must. Here 0 is a perfectly good greatest lower bound, and the entire content of the theorem is that it is not attained.

This is why the completeness axiom for $\mathbb{R}$ is stated in terms of suprema and infima rather than maxima and minima. Had it been phrased with minima it would be false — $(0, \infty)$ is non-empty and bounded below, yet has no least element.

1.9 Remark — Why Analysis Is Written With “For All $\varepsilon > 0$ ”

This theorem is the reason every definition in analysis is phrased with a universally quantified ε . The definition of a limit reads

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x, 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon,$$

and one might reasonably ask why it is not simpler to say “ $|f(x) - L|$ is smaller than the smallest positive real”. The answer is that there is no such thing. Since no single ε is small enough to be definitive, the definition has to quantify over all of them, and the whole ε - δ apparatus follows from that.

The same theorem also underwrites a standard proof move: **if $0 \leq a \leq \varepsilon$ for every $\varepsilon > 0$, then $a = 0$** . If a were positive, taking $\varepsilon = a/2$ gives $a \leq a/2$, the same contradiction as above. Innumerable analysis proofs end this way.

1.10 Remark — On a Computer There Is a Smallest Positive Number

The parallel with the machine-integer remark in Problem 3 is exact. In IEEE 754 double precision there *is* a smallest positive value: the subnormal 5×10^{-324} , roughly 2^{-1074} . Halving it does not produce a smaller positive number, it produces **0** — confirmed by running it, and confirmed too by halving **1.0** repeatedly, which reaches **0** after exactly **1074** steps.

So on a machine, both this theorem and the previous one are false, and for the same structural reason in each case: the representable set is finite, and the witness the proof constructs falls out of it. `int64` fails to be closed under $N + 1$, which wraps around; the doubles fail to be closed under $\varepsilon/2$, which flushes to zero. The theorems are statements about $\mathbb{Z}$ and $\mathbb{R}$, not about their finite approximations — a distinction with real consequences for anyone writing `while (x > 0) x /= 2;`.